Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

Dot .Net Core and Progressive Web Application Practical

Practical – I

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Aim: Create methods add(). multiply(), substract() ,divide() with suitable parameters and call these methods using concept of C# delegate.

Algorithm:

Step1: Start the program.

Step 2: Create a delegate and define calculator methods (Add, Sub, Mul, Div).

Step 3: Create an object of the Calculator class.

Step 4: Assign each method to the delegate and invoke it.

Step 5: Display the results and stop the program.

Code:

```
using System;
delegate void Operation(int a, int b);
2 references
class Calculator
{
    1 reference
    public void Add(int a,int b)
    {
        Console.WriteLine("Addition=" + (a + b));
    }
    1 reference
    public void Sub(int a, int b)
    {
        Console.WriteLine("Subtraction=" + (a - b));
    }
    1 reference
    public void Mul(int a, int b)
    {
        Console.WriteLine("Multiplication=" + (a * b));
    }
    1 reference
    public void Div(int a, int b)
    {
        Console.WriteLine("Division=" + (a / b));
        Console.WriteLine("Afsha 25");
    }
}
```

```
class program
{
    static void Main()
    {
        Calculator c = new Calculator();
        Operation obj;

        obj = c.Add;
        obj(10, 20);

        obj = c.Sub;
        obj(40, 20);


        obj = c.Mul;
        obj(10, 20);

        obj = c.Div;
        obj(60, 20);

    }
}
```

Output:

```
Microsoft Visual Studio Debu
Addition=30
Subtraction=20
Multiplication=200
Division=3
Afsha 25
```

Aim: Write a program using multicast delegate.

Algorithm:

Step1: Start the program.

Step1: Create a delegate and define the methods Welcome(), Learn(), and Thanks().

Step1: Create an object of the Demo class.

Step1: Add all methods to the delegate and invoke it.

Step1: Display the messages and stop the program

Code:

```
public class Program
{
    0 references
    public static void Main(string[] args)
    {
        Demo d = new Demo();
        Message m;

        m = d.Welcome;
        m += d.Learn;
        m += d.Thanks;
        m();

        Console.Write("Afsha 25");

    }
}
```

```
using System;
delegate void Message();
2 references
class Demo
{
    1 reference
    public void Welcome()
    {
        Console.WriteLine("WELCOME...");
    }
    1 reference
    public void Learn()
    {
        Console.WriteLine("Learning C# Delegates");
    }
    1 reference
    public void Thanks()
    {
        Console.WriteLine("Thank You");
    }
}
```

Output:

```
Microsoft Visual Studio Deb
WELCOME...
Learning C# Delegates
Thank You
Afsha 25
```

Aim: Create a class BankAccount with AccountNumber and Balance. Implement property for Balance with validation (Balance cannot be negative).

Algorithm:

Step 1: Start the program.

Step 2: Create the BankAccount class with AccountNumber and Balance properties.

Step 3: Create an object of the BankAccount class.

Step 4: Assign and display the account number and balance.

Step 5: Update the balance, display the details, and stop the program.

Code:

```
using System;
class BankAccount
{
    public int AccountNumber { get; set; }
    private double balance;
    public double Balance
    {
        get
        {
            return balance;
        }
        set
        {
            if (balance >= 0)
                balance = value;
            else
                Console.WriteLine("Balance cannot be negative.");
        }
    }
```

```
    class Program
    {
        static void Main()
        {
            BankAccount b = new BankAccount();
            b.AccountNumber = 101;
            b.Balance = 5000;

            Console.WriteLine("Account Number:" + b.AccountNumber);
            Console.WriteLine("Balance:" + b.Balance);

            b.Balance = -1000;
            Console.WriteLine("Afsha 25");
        }
    }
}
```

Output:

```
Account Number:101
Balance:5000
Afsha 25
```